Hussain Singapuri

AI ENGINEER (LLM, RAG & AGENTS)

+91 9561277305 | Hussainsingapuri24@gmail.com | linkedin.com/in/hussain-singapuri-18437225a | github.com/huss2003 | hussain-singapuri.vercel.app | Pune, India

PROFESSIONAL SUMMARY

AI Engineer with 3 years of professional full-stack experience at a US-based medical-device company, building **AI agents**, a CRM and web apps. Builds **Agentic AI** and **LLM** applications with **RAG** (hybrid BM25 + **pgvector** vector search, embeddings), **tool use** and multi-step agent orchestration, plus **evaluation** harnesses for **hallucination detection** (grounding and unsupported-claim checks), data pipelines (crawl, enrich, index) and **test automation** (end-to-end API tests). Works in **Python** (FastAPI) and TypeScript with **React** front ends, **PostgreSQL**, **REST APIs**, Docker and CI/CD (GitHub Actions). Uses **Claude Code** daily. Claude Certified Developer.

CORE COMPETENCIES

Agentic AI & LLM Applications | RAG & Vector Search (pgvector) | Embeddings & Retrieval | LLM Evaluation & Hallucination Detection | Python & FastAPI | React & TypeScript | PostgreSQL & Data Pipelines | Docker & CI/CD

WORK EXPERIENCE

NeuroNetworks AI Ed

2026 - Present

CTO & Principal Course Architect

Pune, India

- Built and run the website neuronetwrks.com with a Python static-site generator deployed on Vercel, including programmatic SEO
- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)
- Designing Neuromind Practical AI Lab, a browser-based IDE and sandbox with an AI tutor: phased spec, server sizing, engineering team plan and CAPEX/OPEX model

US-based medical-device company

Mar 2023 - Sep 2026

Full Stack Developer

Remote

- Built **AI agents** for internal use
- Built and maintained full-stack web applications for the company
- Built several apps, including a **CRM**
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

PROJECTS

site-to-agent

Open Source

RAG over a website: crawler, embeddings with hybrid BM25 + vector retrieval (pgvector) fused with RRF, cited answers, SSRF-hardened fetching and an eval harness for answer quality.

Next.js, TypeScript, pgvector

multi-agent-research-crew

Open Source

Multi-step agent orchestration with tool use: planner, parallel researchers, writer and verifier on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking.

Python, asyncio, Claude

contractor-ai-ops-dashboard Open Source

Multi-tenant analytics with signed event ingestion, funnel metrics, anomaly alerts and number-checked AI weekly summaries.

Next.js, TypeScript, Recharts, Postgres RLS

PageForge Portfolio project

LLM evaluation and safety controls: AI drafts only from approved facts, a deterministic validator flags unsupported numbers, a reviewer signs off claim by claim.

Next.js 16, React 19, Tailwind v4, Sanity, Supabase, Gemini, PostHog, HubSpot

ai-lead-receptionist Open Source

AI agent for home-service businesses: missed-call text-back, SMS and web chat that qualifies leads and books jobs, owner dashboard.

Next.js, TypeScript, Twilio, Claude, Supabase

nanoserve Open Source

LLM inference engine from scratch: paged KV cache, continuous batching, speculative decoding, OpenAI-compatible streaming API; 3.1x throughput over sequential decoding.

Python, PyTorch, FastAPI

More projects

Orbital Surplus, Personal portfolio with full-stack backend, Nightjar & Co., Safey Inventory, TripSync, Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club

EDUCATION

Bachelor's, Computer Science (AI & Machine Learning) Vishwakarma Institute of Technology, Pune 2022-2025

Pune

CERTIFICATIONS

Claude Certified Developer	Anthropic, Foundations (CCDV-F)
Claude Certified Orchestrator	Anthropic, Foundations (CCAO-F)
Generative AI Leader	Google Cloud
Claude Code in Action	Anthropic
Claude Code 101	Anthropic

SKILLS

AI / LLM: Agentic AI, AI agents, tool use / function calling, multi-agent orchestration, RAG (hybrid BM25 + vector, RRF), evals, prompt engineering, prompt-injection defence, Claude API, Gemini API, LLM inference serving, Claude Code and AI coding agents, hallucination detection (grounding and claim checks)

Languages: TypeScript, JavaScript, Python, SQL, Dart, HTML, CSS

Data: PostgreSQL, Neon, Supabase (RLS, Edge Functions), pgvector, PGlite, SQLite, Sanity CMS, data pipelines (crawl, enrich, index)

Back end: Node.js, REST APIs, FastAPI, asyncio, webhooks (HMAC signing), auth (scrypt, sessions, throttling), Stripe Checkout and refunds, Resend, test automation (end-to-end API tests)

Front end: React 19, Next.js, TypeScript, JavaScript, Tailwind CSS, shadcn/ui, TanStack Start / Query, GSAP, Lenis, three.js, WebGL, Canvas 2D, responsive and accessible UI, motion and interaction design

DevOps: Docker, GitHub Actions, Vercel

Security: mobile app penetration testing, CSP and security headers, SSRF hardening, input validation

Automation & integrations: workflow automation, CRM integrations (HubSpot), PostHog, Twilio SMS, web crawling / scraping

Mobile: Flutter, Riverpod, Firebase **Design:** UI/UX design, design systems, technical SEO, JSON-LD structured data, programmatic SEO